

Aurélien Cécille

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Education

Ph.D. Candidate in Computer Science

LIRIS & Visual Behavior May 2023 - Present
Thesis: 'Monocular depth: self-supervised learning integrating scene structure'

Master's Degree in Computer Science

University of Technology of Troyes 2017 - 2022

Work Experience

Research Engineer

Visual Behavior Oct. 2022 - May 2023

- Research, training, and evaluation of deep learning models for robotic and intelligent vehicle vision

AI Engineer Intern

Orange Feb. 2022 - Jul. 2022

- Detection and segmentation of fiber optic cables in images
- Few-shot learning methods based on meta-learning
- Leveraging latent representations from generative models (GANs)

AI Engineer Intern

Visual Behavior Sept. 2020 - Feb. 2021

- Self-supervised learning for depth estimation, optical flow, and egomotion estimation
- Environment reconstruction from bird's-eye view

Projects

Conditional Logo Generation

- Logo generation using StyleGAN2
- Conditioning the generator with latent-space clustering from a VAE
- Flask interface with color, shape, and category controls

Reinforcement Learning Agent

- Custom grid-world environment in Unity ML-Agents linked with OpenAI Gym
- Implementation in JAX for parallelization and optimized hyperparameter search

Research Interests

Computer Vision Deep Learning

Self-Supervised Learning

Publications

Towards Sharper Object Boundaries in Self-Supervised Depth Estimation

Aurélien Cécille, Stefan Duffner, Franck Davoine, Rémi Agier & Thibault Neveu
BMVC oral, November 27, 2025, Sheffield, United Kingdom

GroCo: Ground Constraint for Metric Self-Supervised Monocular Depth

Aurélien Cécille, Stefan Duffner, Franck Davoine, Thibault Neveu & Rémi Agier
ECCV 2024, October 2024, Milan, Italy

Technical Skills

Deep Learning

PyTorch, JAX, TensorFlow

Machine Learning

Scikit-learn, Pandas

Data Visualization

D3.js, Python, R, Tableau, Power BI

Web Applications

Flask, React, Vue.js

Programming Languages

Python, Java, C

Tools & Databases

Docker, SQL, NoSQL

Languages

English : Fluent (C2 level)
French : Native language